



UNIVERSITÀ DEGLI STUDI DI PADOVA

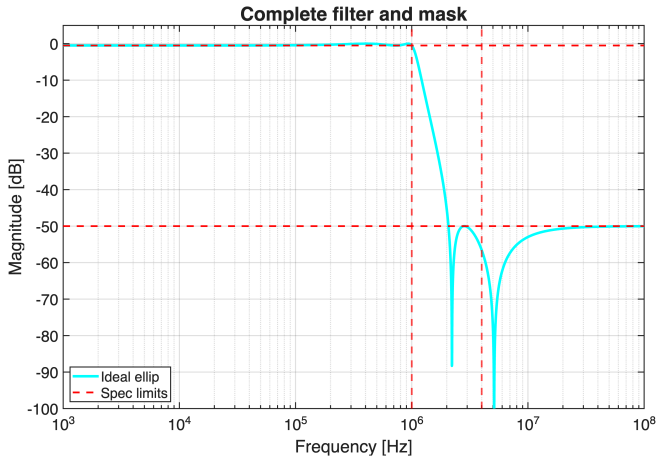
Lab 1: 4th-order elliptic low-pass filter

Integrated Circuits for Signal Processing

Lorenzo Tomba 2188353, Peter Collavino 2188355
Gregorio Massella 2165994, Alveena Magsi 2194853

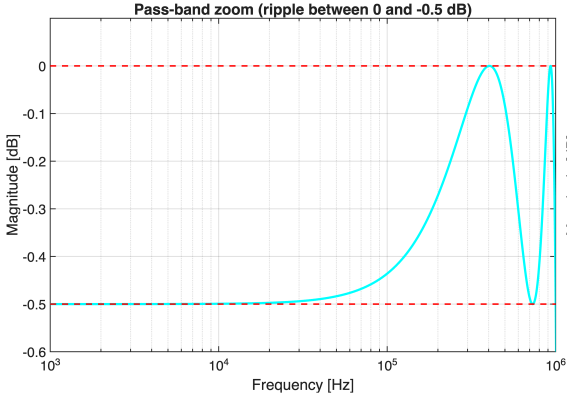
2 November 2026

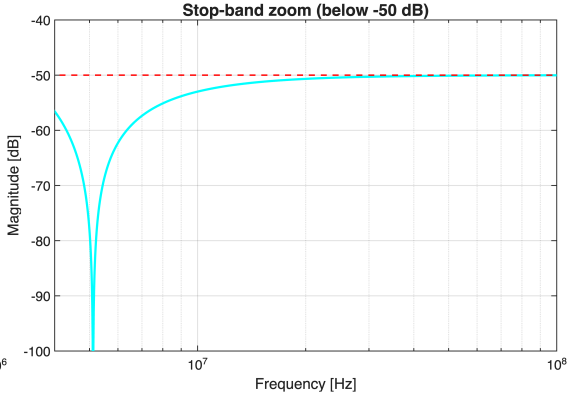
Matlab Mathematical Result & Specifications



- 4th-order low-pass elliptic filter, ellipord gives minimumm order $N = 3$
- Pass band edge $f_p = 1$ MHz, stop pand edge $f_s = 4$ MHz
- In-band ripple ≤ 0.5 dB, stop-band attenuation ≥ 50 dB
- Output noise $< 120\ \mu V_{\mathrm{rms}}$

Pass Band & Stop Band Zoom





Biquad Decomposition

- The overall filter transfer function can be decomposed into two separate second-order transfer functions.
- For each transfer function, we determine k_0, k_1, k_2, Q, and ω_0. Biquad 1 has the lower Q and comes first in the cascade, while Biquad 2 has higher Q and all the DC gain of the total transfer function

$$H(s) = \frac{k_2 s^2 + k_1 s + k_0}{\frac{s^2}{\omega_0^2} + \frac{s}{Q\omega_0} + 1}$$

Biquad 1

Parameter	Value
ω_0 [rad/s]	4.0039×10^6
Q	7.2264×10^{-1}
k_0	1.0000
k_1	0.0000
k_2	9.6813×10^{-16}

Biquad 2

Parameter	Value
ω_0 [rad/s]	6.4771×10^6
Q	3.3831
k_0	9.4406×10^{-1}
k_1	0.0000
k_2	4.8566×10^{-15}

Component Sizing from the Noise Specification

- The extracted parameters constrain the RC time constants. Component values must also meet the noise limit; deriving each noise-source-to-output transfer function is a long procedure, so we use a simpler estimate.
- Output noise power scales as $16kT/C$. With equal capacitors, a noise check lets us rescale all capacitances proportionally to the noise limit of the specifications.

$$C_a = C_b = 16 \cdot \frac{kT}{V^2_{\mathrm{noise,RMS}}} \simeq 4.6\,\mathrm{pF}$$

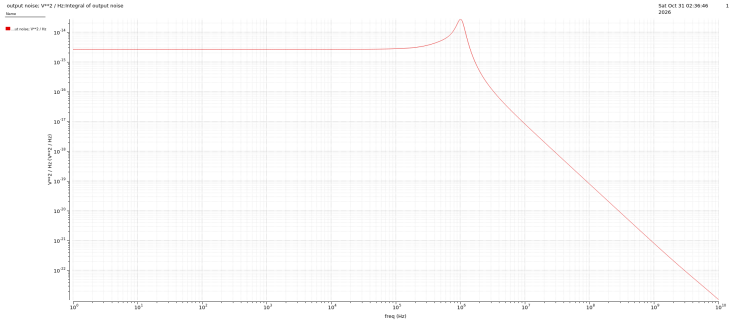
Biquad 1

Component	Initial	After noise check
$C_a = C_b$	4.6 pF	9.2 pF
R_1	39.2358 kΩ	19.6179 kΩ
R_3	54.2948 kΩ	27.1474 kΩ
R_4	54.2948 kΩ	27.1474 kΩ
R_5	−54.2948 kΩ	−27.1474 kΩ
C_4	−0.0714 pF	−0.1428 pF

Biquad 2

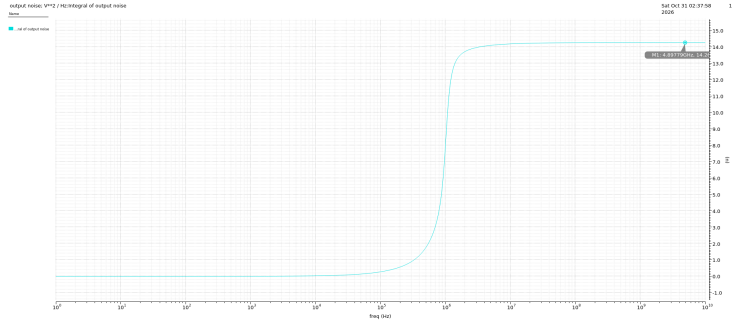
Component	Initial	After noise check
$C_a = C_b$	4.6 pF	9.2 pF
R_1	113.5463 kΩ	56.7732 kΩ
R_3	33.5629 kΩ	16.7814 kΩ
R_4	35.5516 kΩ	17.7758 kΩ
R_5	−33.5629 kΩ	−16.7814 kΩ
C_4	−0.9372 pF	−1.8745 pF

Power Spectral Density of Noise



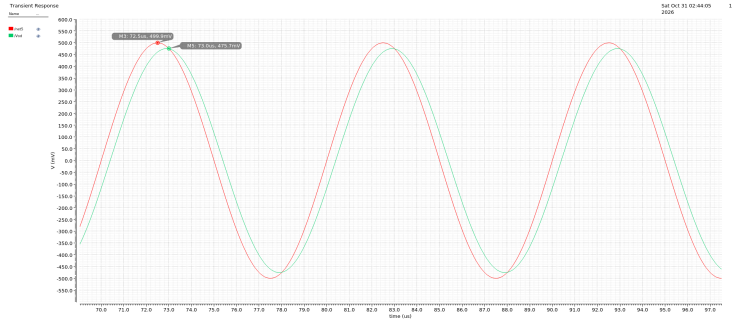
- The plotted PSD is integrated from 1 Hz to 10 GHz to obtain the noise power of interest.
- The total PSD is the sum of the contributions from the resistor noise sources, each weighted by its corresponding transfer function.

Integrated Noise Power



- The RMS noise voltage is obtained by taking the square root of the integrated noise power.
- The initial design yields about twice the target noise power. As discussed, doubling the capacitances reduces it to the target level.
- The final output noise is 119.4 μV_{rms}.

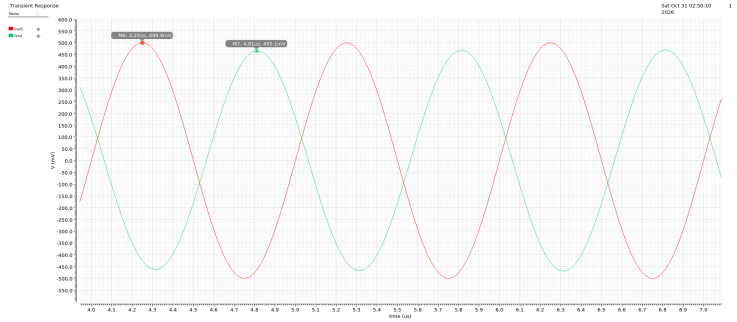
Transient Response 100 kHz



- The plot shows the input and output sinusoids together for a quick comparison.
- The transfer function introduces a small attenuation and a phase shift in the output sinusoid.
- The attenuation remains within the specification:

$$A_{1,\mathrm{dB}} = 20\log_{10}\left(\frac{A_{\mathrm{out}}}{A_{\mathrm{in}}}\right) = -0.43\,\mathrm{dB}$$

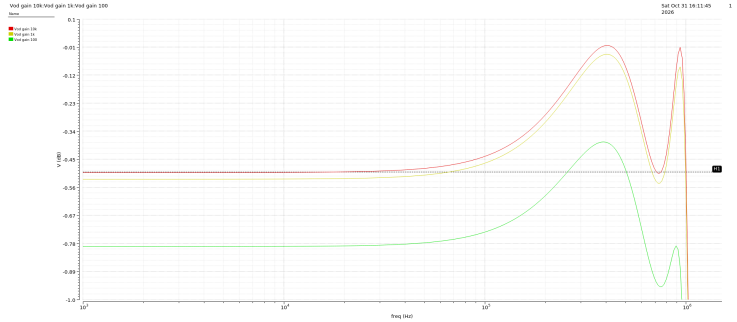
Transient Response 1 MHz



- At the bandwidth limit, the output sinusoid exhibits a large phase shift.
- The attenuation is slightly outside the specification:

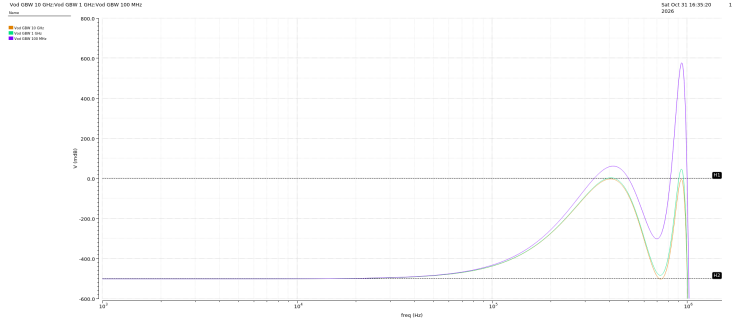
$$A_{2,\mathrm{dB}} = 20\log_{10}\left(\frac{A_{\mathrm{out}}}{A_{\mathrm{in}}}\right) = -0.63\,\mathrm{dB}$$

Limited OPAMP gain effects



- Reducing the gain causes a drop before the edge, pushing the response outside the specifications.
- Tested gains: 100, 1k, and 10k.

Limited OPAMP bandwidth effects



- Reducing the bandwidth creates a higher peak at the pass-band limit.
- Tested bandwidths: 100 MHz, 1 GHz, and 10 GHz.